

Title: Predictive Modeling and Risk Scoring for Customer Churn in Retail Banking

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Abstract

Customer churn poses a significant challenge for retail banks, directly affecting customer lifetime value, revenue stability, and competitiveness. This study develops a predictive churn intelligence system using a dataset of 10,000 customers with demographic, financial, and behavioral attributes. After preprocessing and feature engineering, multiple machine learning models were trained, including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, LightGBM, and XGBoost. Results indicate that Gradient Boosting and LightGBM achieved the best balance between accuracy and recall, with ROC-AUC scores above 0.85. Feature importance analysis revealed that age, number of products, and customer activity status were the strongest predictors of churn. These findings provide actionable insights for proactive customer retention strategies.

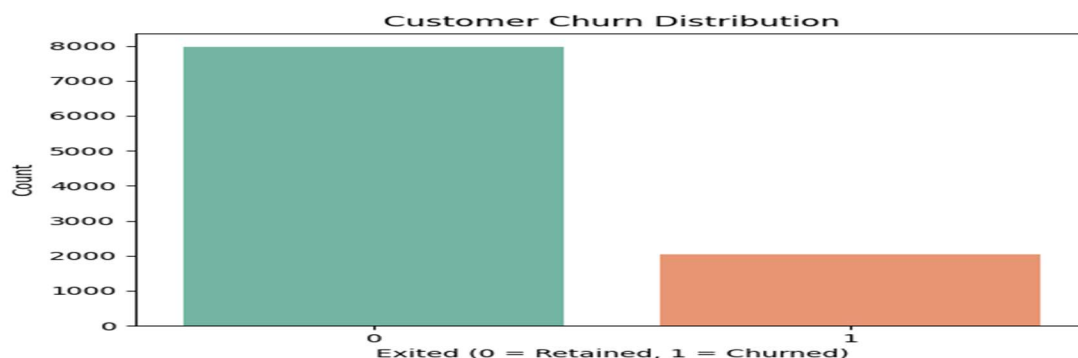
Introduction

Customer churn is a critical issue in retail banking, where retaining existing customers is often more cost-effective than acquiring new ones. Traditional churn analysis explains past attrition but fails to provide proactive insights. Predictive modeling enables banks to identify at-risk customers before they leave, allowing targeted interventions. This study aims to build and evaluate machine learning models for churn prediction, assess feature importance, and provide a framework for risk scoring.

Methods

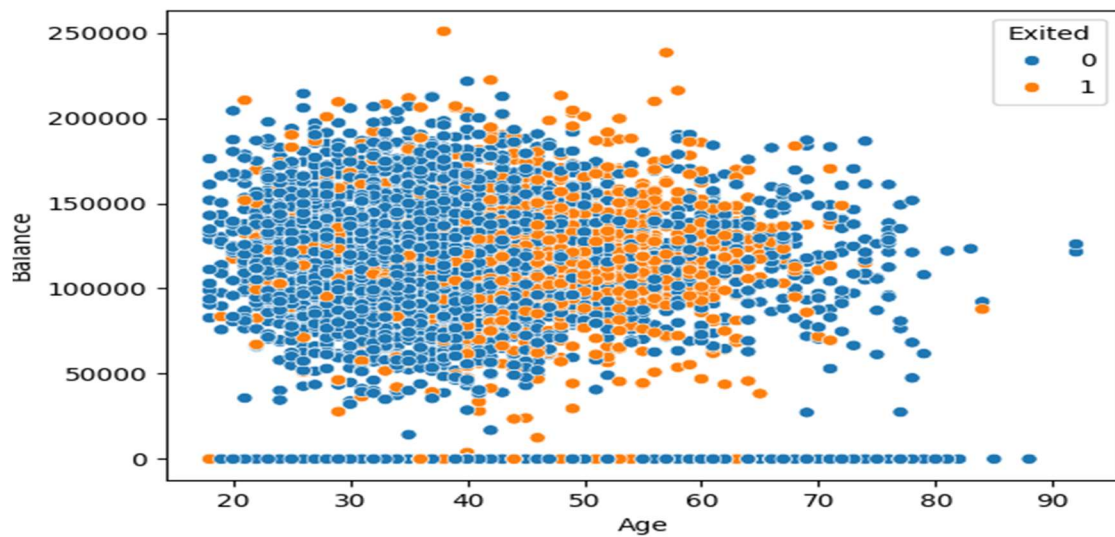
Dataset

The dataset consists of 10,000 customer records with attributes including **CreditScore**, **Geography**, **Gender**, **Age**, **Tenure**, **Balance**, **NumOfProducts**, **HasCrCard**, **IsActiveMember**, **EstimatedSalary**, and **Exited (churn indicator)**. No missing values were observed. The churn distribution was imbalanced: 7,963 retained customers (Exited = 0) and 2,037 churned customers (Exited = 1).

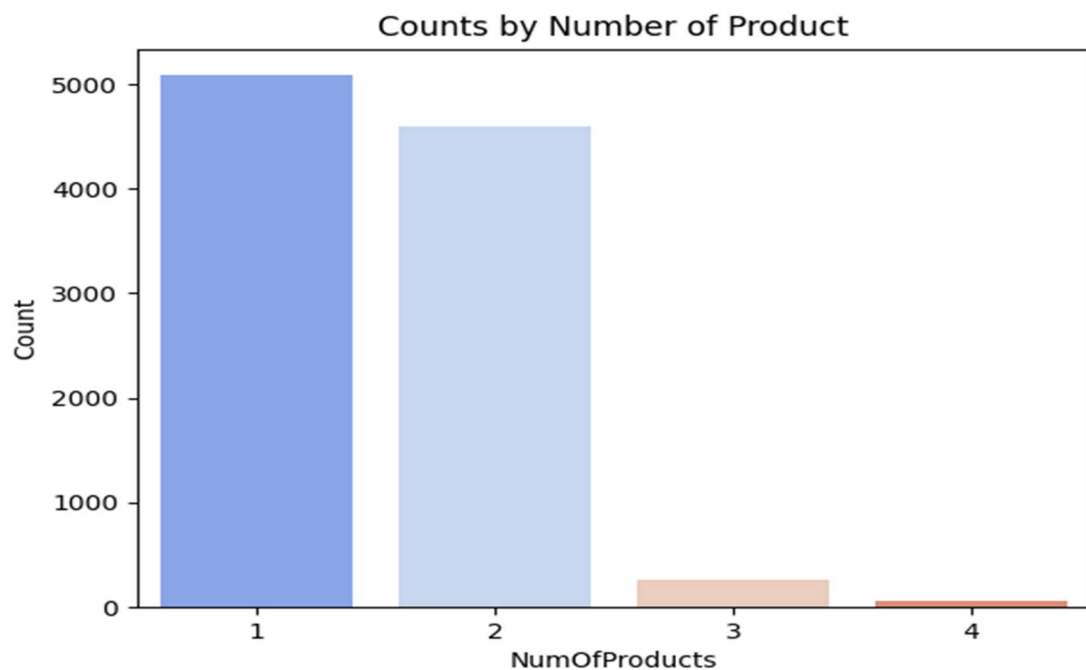


Preprocessing

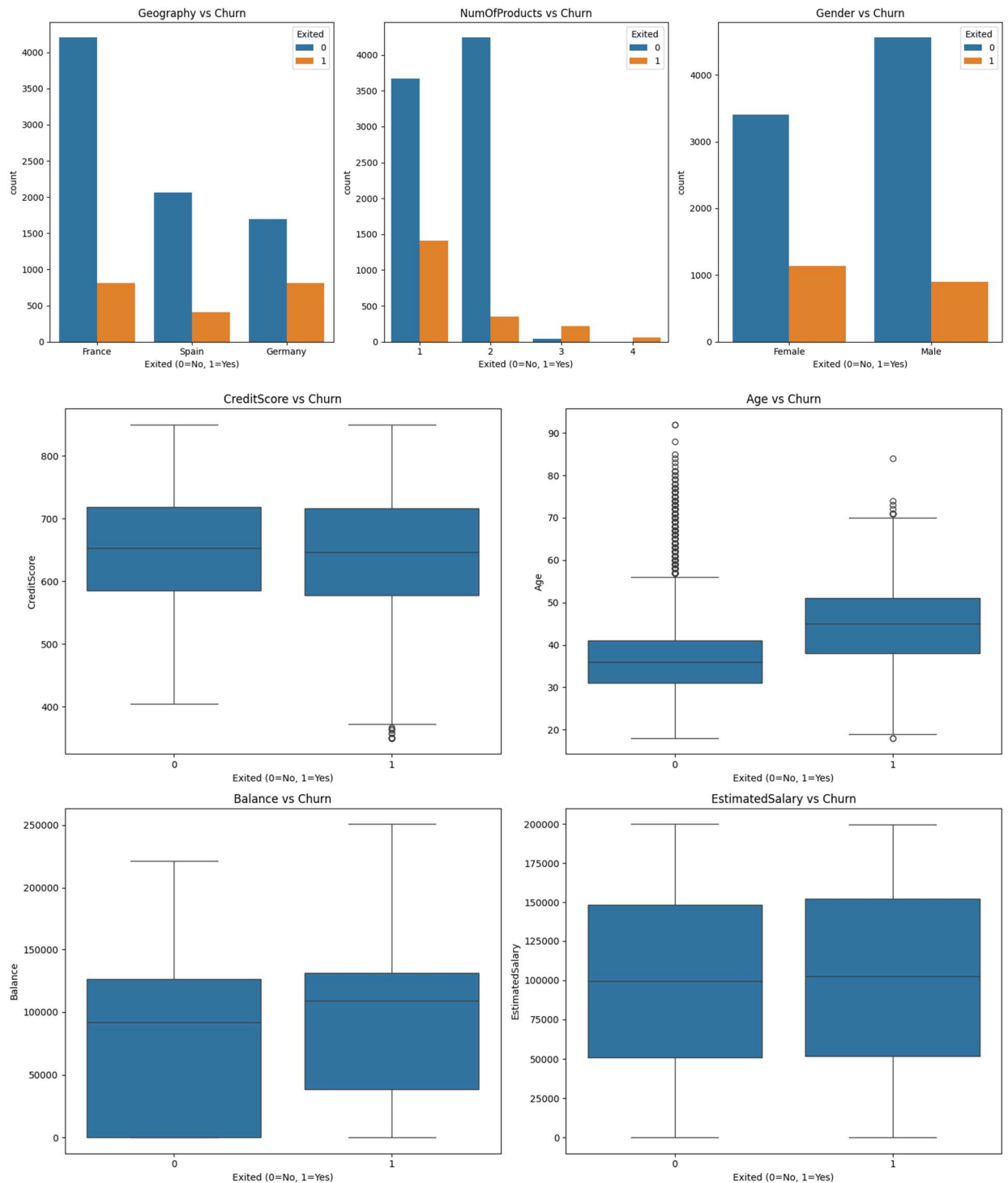
- **Feature Engineering:** Derived variables included BalanceSalaryRatio, ProductDensity, EngagementProduct, and AgeTenureInteraction.
- **Train-Test Split:** 80% training (8,000 samples) and 20% testing (2,000 samples).
- **Encoding:** Categorical variables (Geography, Gender) were one-hot encoded.
- **Scaling:** Continuous variables were standardized.



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Models

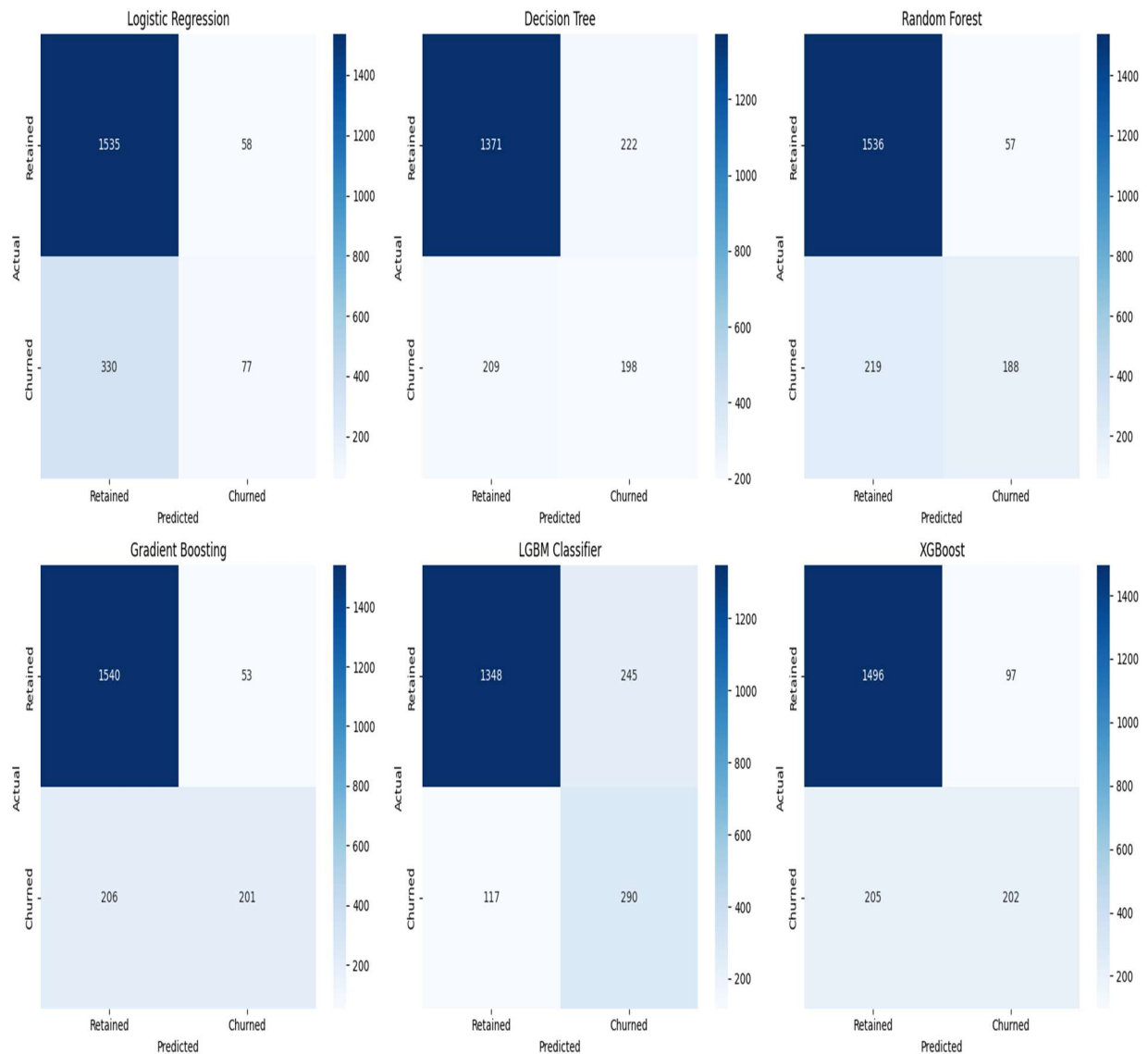
Six models were trained and evaluated:

- Logistic Regression

- Decision Tree
- Random Forest
- Gradient Boosting
- LightGBM
- XGBoost

Evaluation Metrics

Models were assessed using **Accuracy, Precision, Recall, F1-Score, and ROC-AUC**. Confusion matrices were also analyzed to understand misclassification patterns.



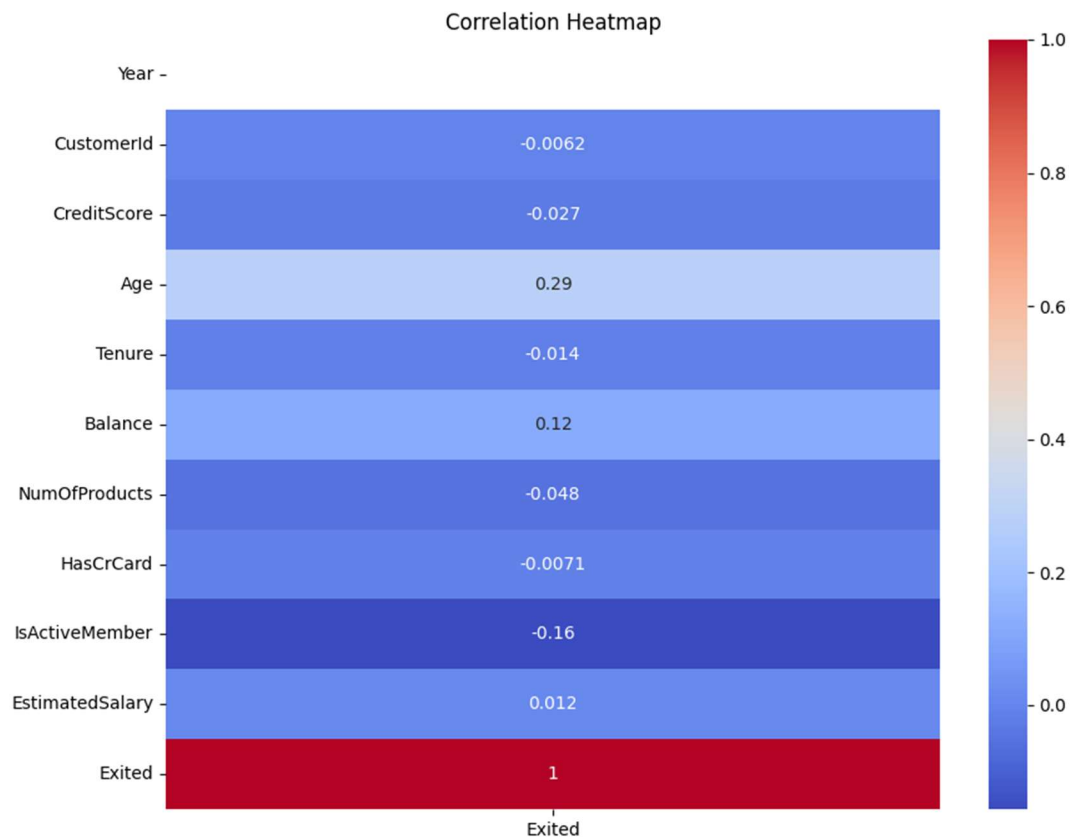
Results

Descriptive Statistics

- Average age: 38.9 years (SD = 10.5).
- Average credit score: 650.5 (SD = 96.6).
- Average balance: 76,485.9 units.
- Gender distribution: 54.6% male, 45.4% female.
- Geography distribution: France (50%), Germany (25%), Spain (25%).

Correlation Analysis

- Age showed the strongest positive correlation with churn ($r = 0.29$).
- IsActiveMember was negatively correlated ($r = -0.16$).

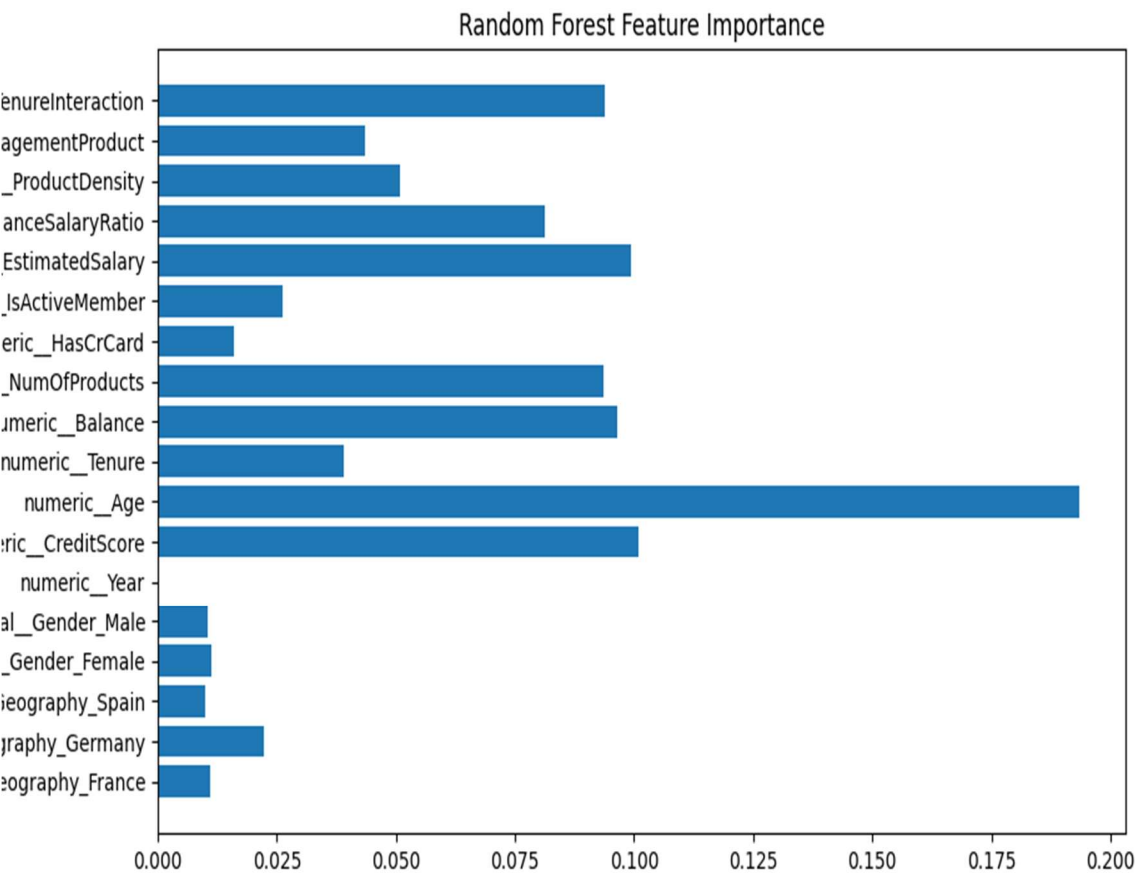


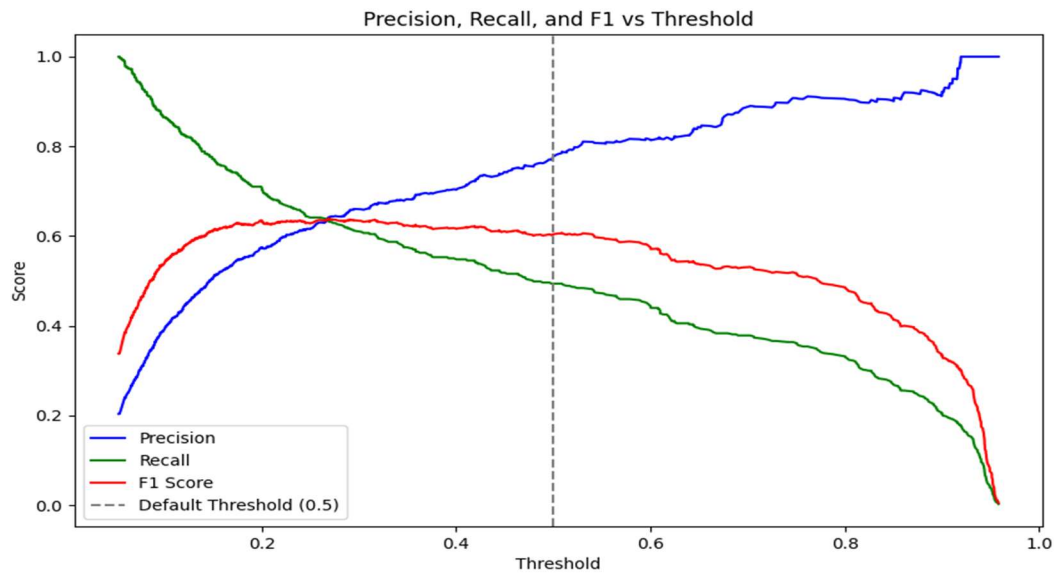
- Balance had a moderate positive correlation ($r = 0.12$).

Model Performance

LightGBM achieved the **highest recall (0.71)**, making it most effective for identifying churners, while Gradient Boosting achieved the **highest overall accuracy (0.87)**.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.806	0.57	0.19	0.28	0.77
Decision Tree	0.785	0.47	0.49	0.48	0.67
Random Forest	0.862	0.77	0.46	0.58	0.85
Gradient Boosting	0.871	0.79	0.49	0.61	0.87
LightGBM	0.819	0.54	0.71	0.62	0.86
XGBoost	0.849	0.68	0.50	0.57	0.83





Discussion

The results highlight the trade-off between precision and recall in churn prediction. Logistic Regression underperformed due to its linear assumptions, while ensemble methods (Random Forest, Gradient Boosting, LightGBM, XGBoost) captured complex relationships more effectively. LightGBM's high recall suggests it is particularly useful for minimizing false negatives, which is critical in churn prevention. Feature importance analysis confirmed that **Age, NumOfProducts, and IsActiveMember** are the most influential predictors, aligning with business intuition: older customers and inactive members are more likely to churn.

Conclusion

This study demonstrates the effectiveness of machine learning in predicting customer churn in retail banking. Gradient Boosting and LightGBM models provide strong predictive performance, with LightGBM excelling in recall. Banks can leverage these models to assign churn risk scores and design targeted retention strategies. Future work should explore explainable AI techniques (e.g., SHAP values) to enhance interpretability and integrate churn prediction into real-time customer engagement systems.

References

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